A Deep Learning Based Implementation for Self-Driving Car

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2. Perception
   - Traffic Signs Detection
   - Lane Detection
   - Dynamic Obstacle Detection
3. Motion Planning
   - Road planning
   - Sign Planning
   - Dynamic Obstacle Planning
4. Navigation

Related Researches

- The main points we discussed above are the self-driving car consisting of three modules:
  - Perception
  - Motion Planning
  - Navigation

Perception

- Three problems need to be solved:
  - Traffic signs
  - Lane detection
  - Dynamic obstacles

Traffic Signs Detection

- Purpose: How do self-driving car learn to recognize object?
- Object detection one stage methods: SSD (Single-shot detection), Yolo (You Only Look Once) SOTA.

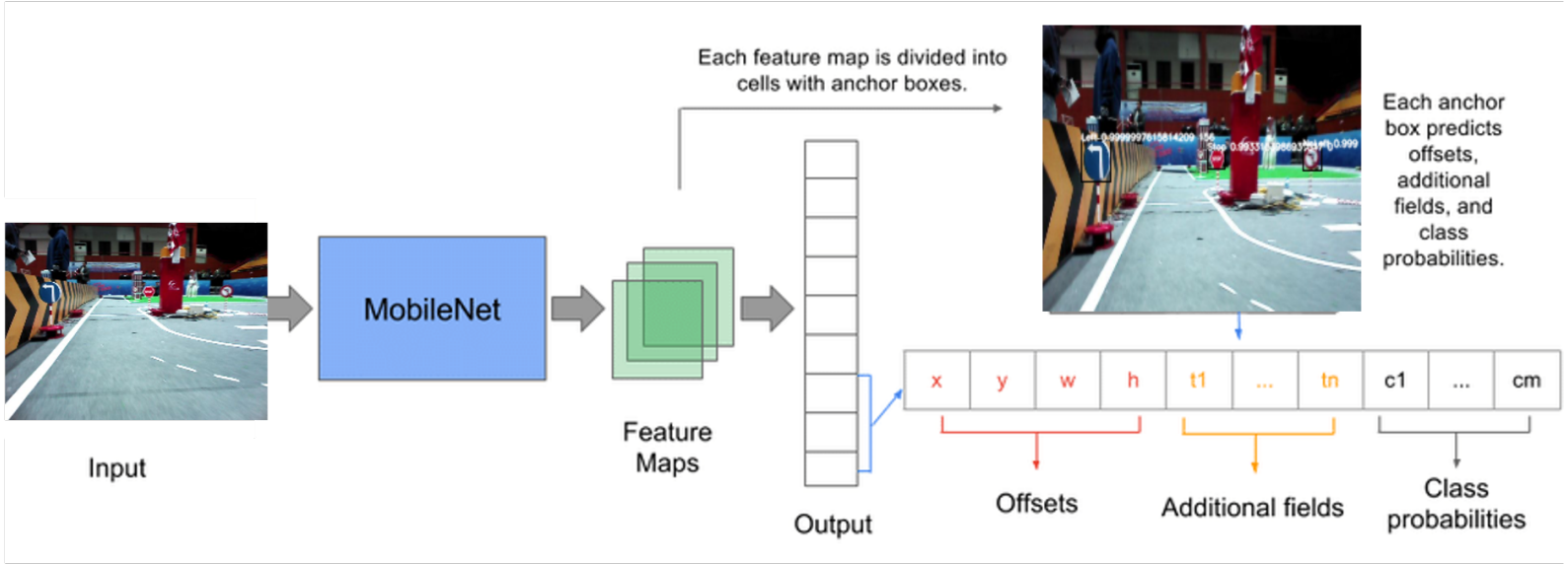


Figure: Digital Race Competition.

- We implement SSD 7 layers (lightweight to realtime - 60 fps).

Lane Detection - Road Segmentation

- Purpose: How do driverless cars work?

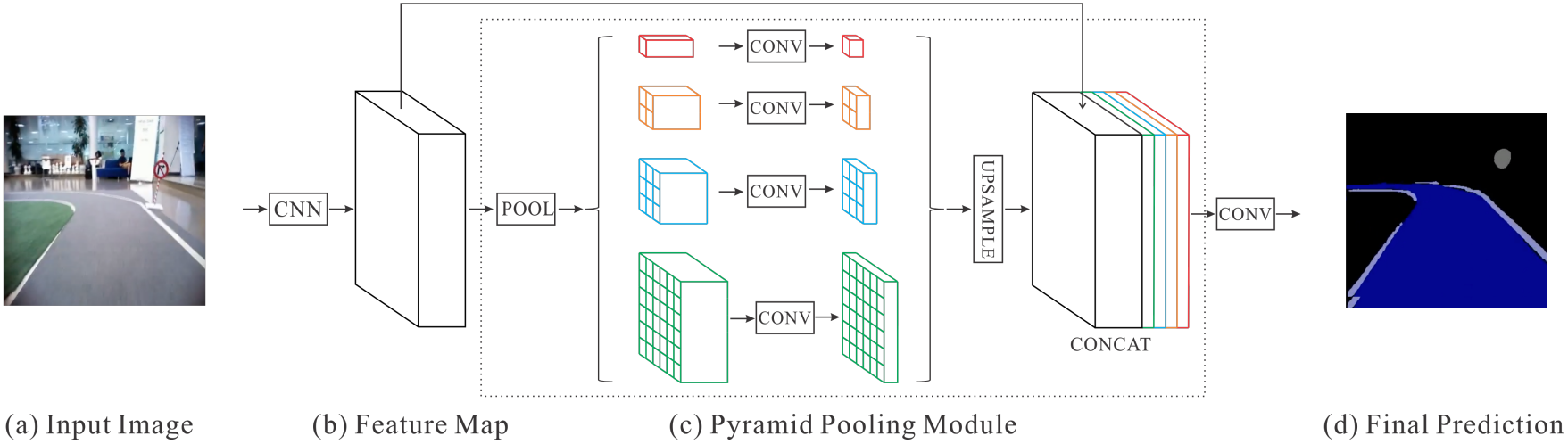


Figure: PSPNet architecture for semantic segmentation task.

- We propose to implement semantic segmentation task to predict angle of car.

Dynamic Obstacle Detection

- A new cluster is started whose first point is p_n

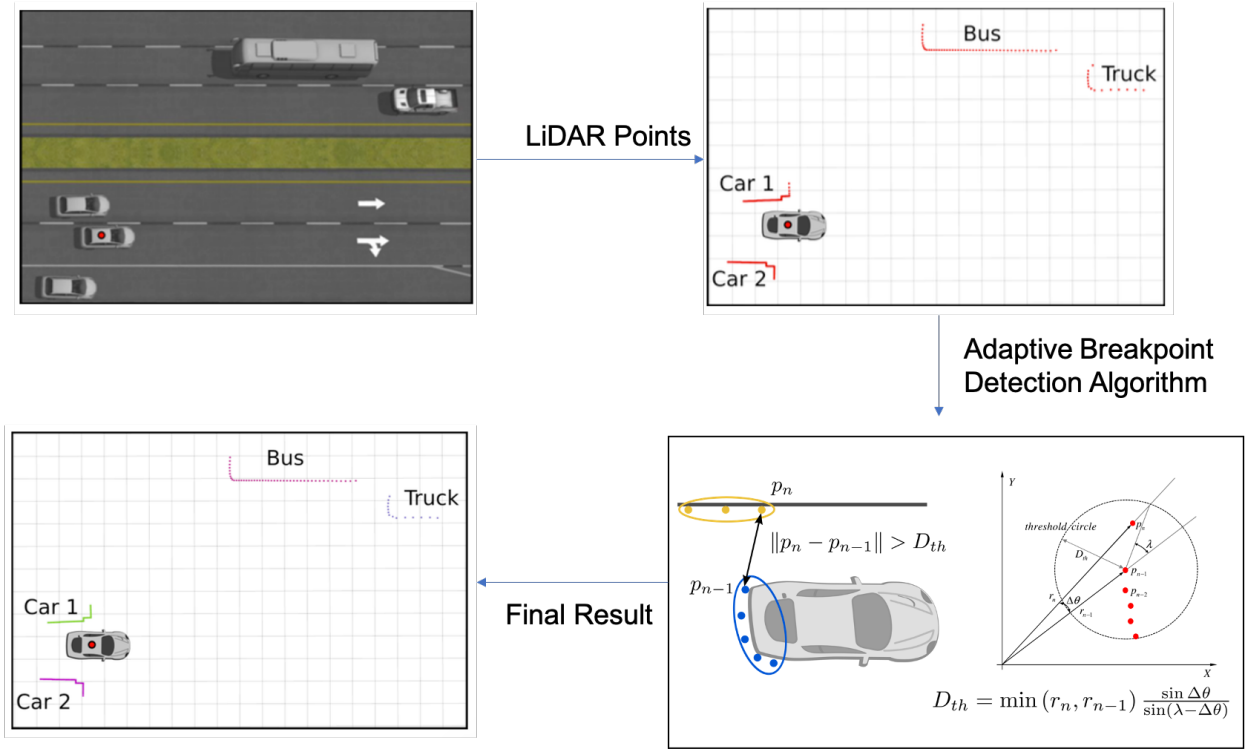


Figure: Adaptive Breakpoint Detection Algorithm

- Find separate clusters if $||p_n - p_{n-1} > d_{th}||$. Then update d_{th} and find all points $< d_{th}$.

Data Association

- The clusters are tracked between time frames by a Nearest Neighbour data association scheme, with a Euclidean distance criterion.

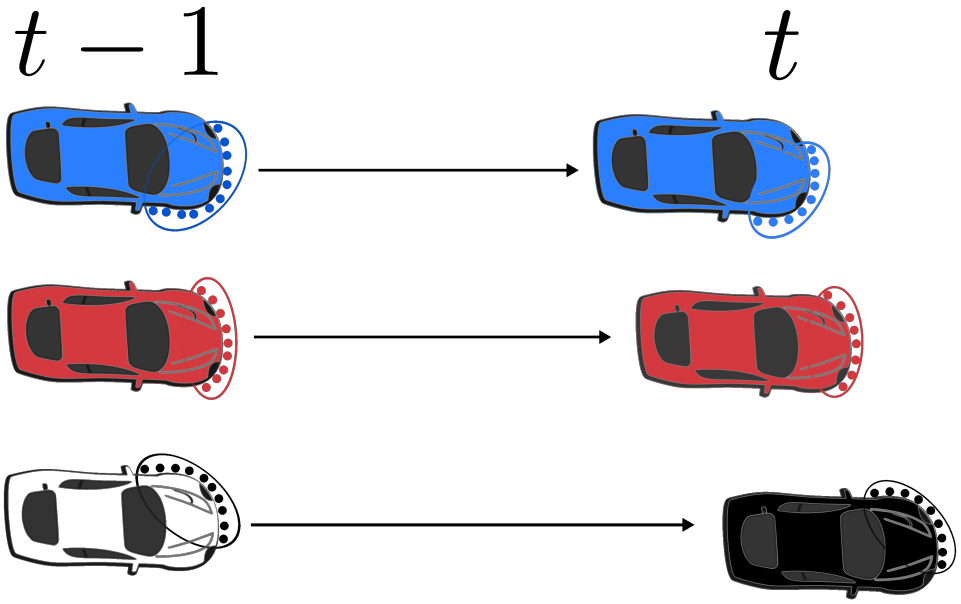


Figure: Global Nearest Neighbour Algorithm

Road path planning

- We have three modes: Left-mode, straight-mode and right-mode.

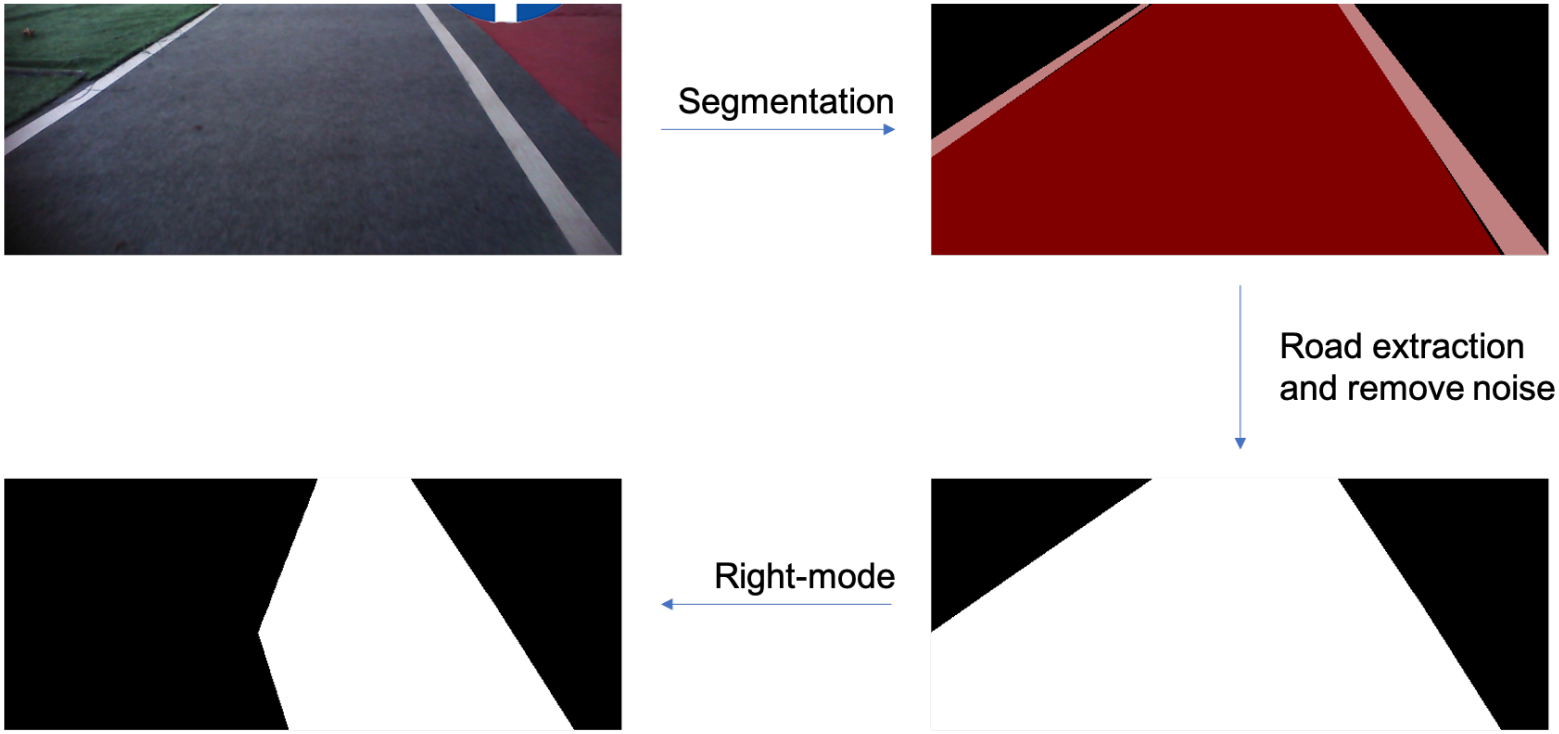


Figure: Road Path Planning

- In road extraction step, we remove noise from segmentation prediction to obtain pure road.

Traffic Sign Path Planning

- Self-driving Car path planning is based on vertical road densities r_v and horizontal line densities l_h.

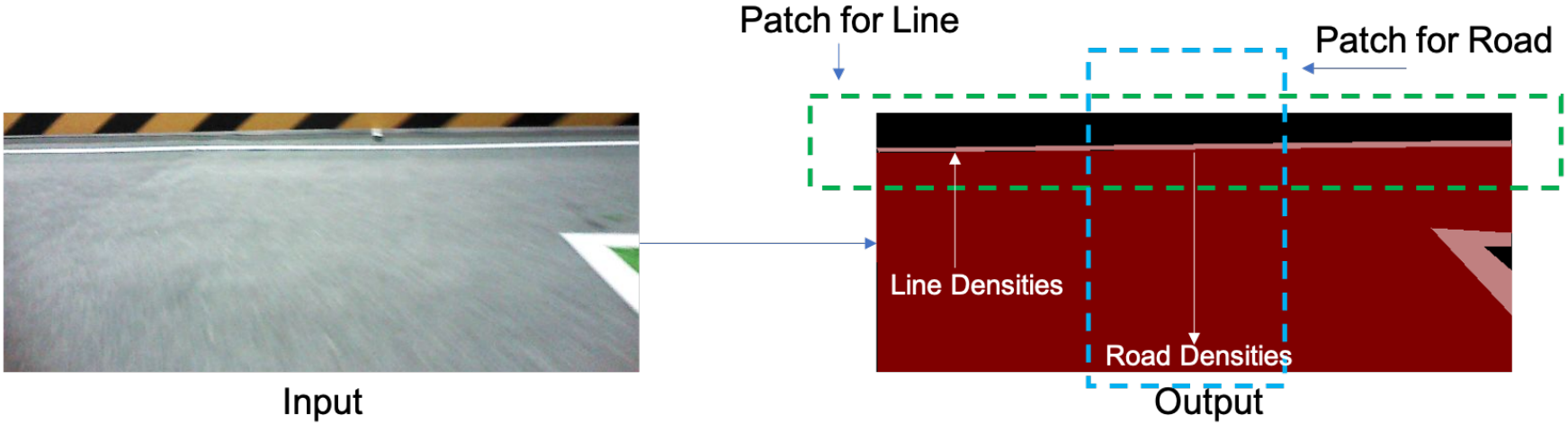
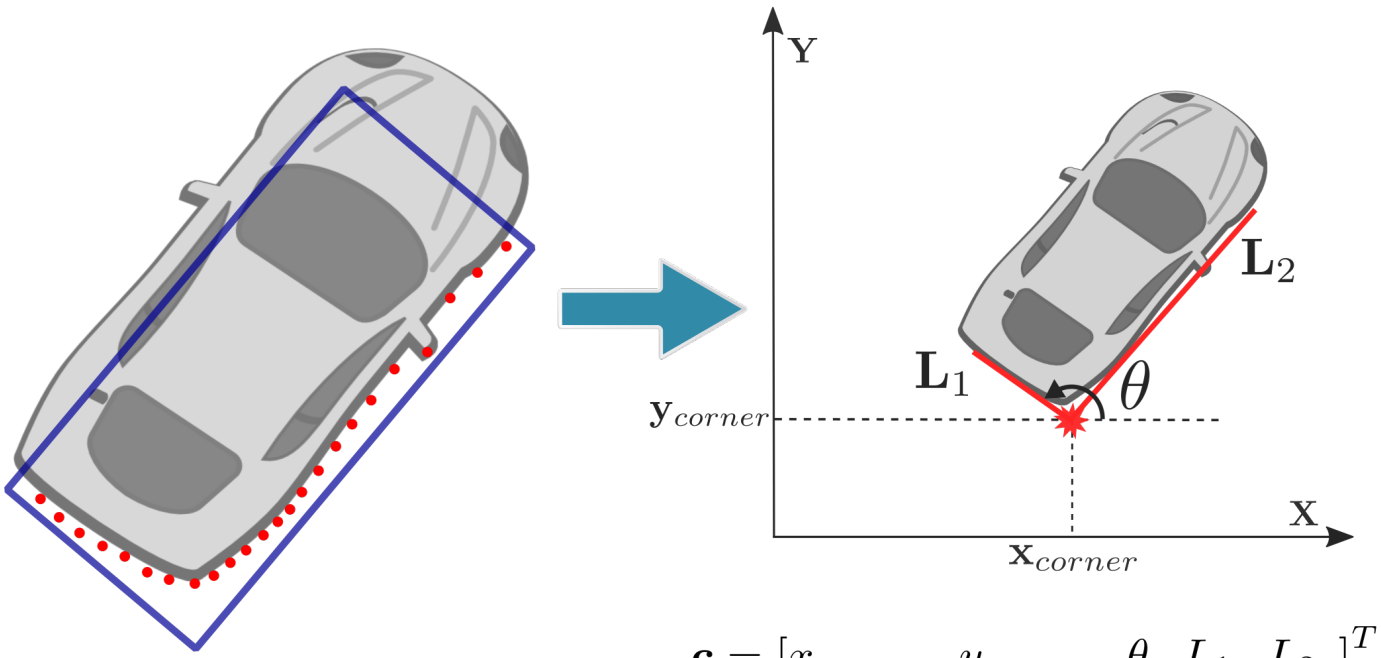


Figure: Global Nearest Neighbour Algorithm

- We cut a small patch p_r is equal 1/3 width of image and p_l is equal 1/3 height of image.
- We set threshold for r_v is lower than 70% and l_h is greater than 80% in output size (144 x 144):

Dynamic Obstacle Planning

- Based on width box and length box, we classify dynamic obstacle is the car.



$$\mathbf{c} = [x_{corner},\ y_{corner},\ \theta,\ L_1,\ L_2,]^T$$

Figure: Find car obstacle by closest corner point.

Dynamic Obstacle Planning

- Tracking ID of cluster (car obstacle) by Kalman Filter Uncertainty.
- Rotate to left-mode and comeback to right-mode when LiDAR can't find any clusters in threshold region.

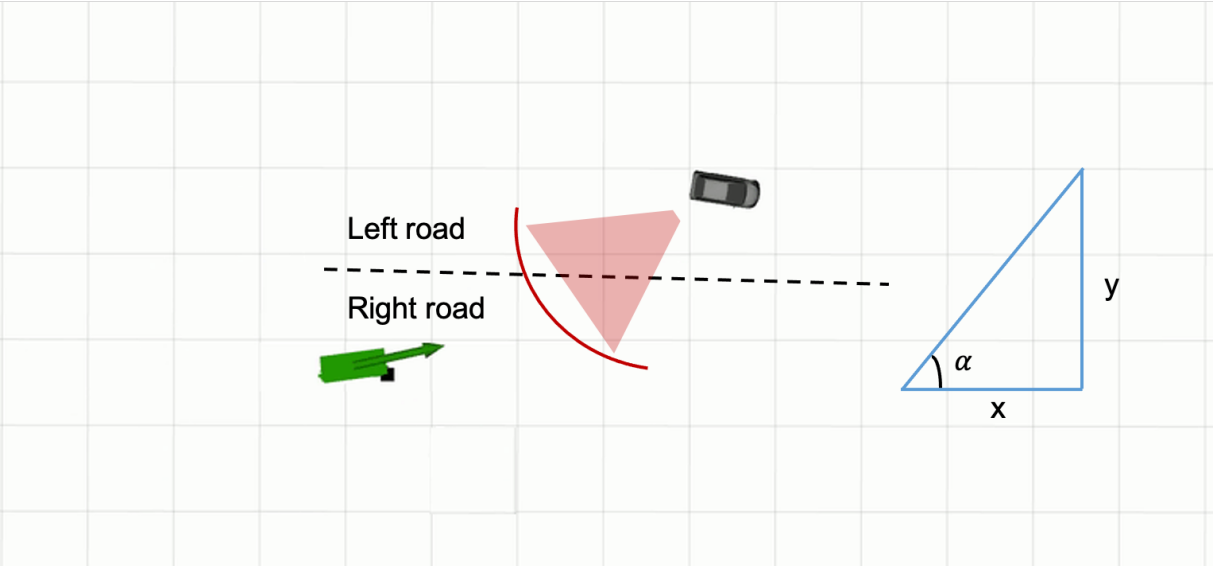


Figure: Scan threshold region.

- x,y is the coordinates of closest corner point, we set threshold for $\alpha \in (\frac{\pi}{6}, \frac{\pi}{2})$.

Compute angle of self-drving car

- Predict angle is based on computing centroid of road extraction.

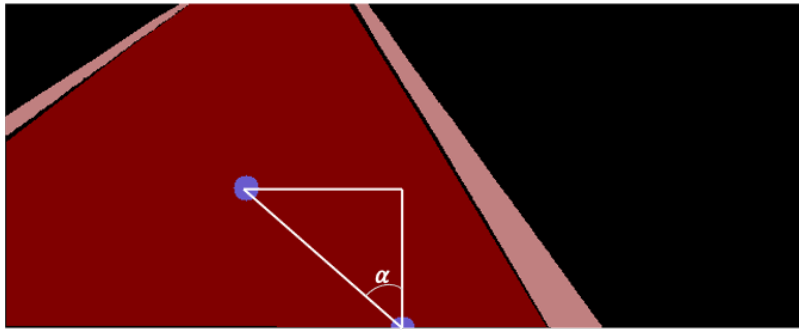


Figure: Compute angle.

- Angle is computed by formula: $\alpha = tan^{-1}(\frac{b}{a})$ with $b = x_{road} - x_{frame}$ and $a = y_{road} - y_{frame}$.